

Explainability of SMOTE Based Oversampling for Imbalanced Dataset Problems

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Abstract—Since the advent of Artificial Intelligence (AI), the problem of imbalanced datasets and the lack of interpretability of complex AI models has been a matter of concern for the research community. These datasets contain a very low proportion of one class (minority class) and very large proportion of another class (majority class). Even though the quantitative representation is less for minority class they have high qualitative importance as the cost associated in case of misclassification in these domains is very high. The paper presents a novel solution to deal with the issue of imbalanced dataset by using the proven method of resampling Synthetic Minority Oversampling Technique (SMOTE). Further, the interpretability of such an approach is demonstrated by some powerful eXplainable AI (XAI) techniques such as LRP, SHAP and LIME. In this paper state-of-art models like Deep Learning and Boosting classifiers were trained to classify fraud instances with high accuracy and proved to be reliable by producing explanations for their predicted instances. The results of confusion matrices and explanations showcase excellent performance and reliability of the models.

Keywords—imbalanced dataset; XAI; SMOTE; MLP; XG-boost; SHAP; LRP; LIME

I. INTRODUCTION

While training classification models, machine learning practitioners often face problems of having unequal distributions of various classes. Such imbalanced datasets are often found in use cases of fraud detection, network intrusion detection, medical diagnosis, oils spills detection in satellite images and condition monitoring. The minority class distribution in such datasets is generally very less comprising 2-3% of total samples. This causes the machine learning models to get biased towards the majority class leading to a classifier with a high classification accuracy, contributed by the majority class but has a poor ability to classify the minority class. The minority class in such datasets occurs rarely whose incorrect classification often leads to catastrophic consequences.

Further the classification is not interpretable as most classifiers behave as black boxes. In order to resolve these challenges, the purpose of this research mainly focuses on improving the reliability of the imbalanced classifiers and making the predictions more explainable with the help of new techniques of Explainable AI. The dataset used was given by [1] and consists of over 6 million samples.

The imbalanced dataset problem can be solved either by reducing the size of the majority class (undersampling) or by increasing the samples of the minority class

(oversampling) by creating samples similar to the existing ones to create an equally proportioned dataset. Resampling the dataset to balance the distribution of each class is the most widely used technique for dealing with imbalanced datasets. Prominently used resampling techniques include undersampling and oversampling. [2] and [3] show that oversampling techniques tend to be more superior than undersampling as the latter tend to lose potential data. Oversampling techniques could be random as well as informative. Various techniques that are used for dealing with imbalanced datasets are given in [4] and [5]. This study mentions the wide use of oversampling techniques like Mega-trend Diffusion Function (MTDF), Synthetic Minority Oversampling Technique (SMOTE), Adaptive Synthetic Sampling approach (ADASYN) [4], Majority Weighted Minority Oversampling Technique (MWMOTE) [6] being the best of these. [7] mentions the comparative study of SMOTE and MTDF. [8] demonstrates the use of SMOTEBoost as being the superior algorithm to deal with the class imbalance problem. Ripple- SMOTE, a hybrid approach involving data manipulation and algorithm has been proposed by [9].

SMOTE, first introduced by [10] is the most widely used technique for resampling the imbalanced datasets. Work by [11] shows that SMOTE has the potential to reduce the class imbalance for low dimensional feature space. [12] has also shown outstanding results by adopting active learning based SMOTE, where SVM is introduced in the learning algorithm of SMOTE for class-imbalance learning. [13] introduced an ensemble based technique of oversampling called DeepBalance.

The use cases where imbalance datasets are found (like fraud detection) are sensitive areas demanding justification. In these areas highly accurate models do not suffice, the explanation for each prediction is equally important. In many instances even if the model provides high accuracy the practitioner finds itself in a dilemma, whether the model used is reliable or not. For this we propose the use of newly introduced techniques of Explainable AI for justifying the predictions of our model. This concept is currently gaining momentum in the research world as it is termed as the “third wave” of AI by DARPA [14]. This “third wave” in AI called Contextual adaptation, focuses on creating algorithms which would provide models with higher accuracy and explainability than present models.

The concept of Explainable AI (XAI) is one of the most trending topics of research in the field of machine learning. The concept is not totally new as the works of [15] in 2002 and many others successfully show the applicability of interpretable neural network in their respective domains. Many algorithms are now being used by researchers to make the existing AI models more interpretable. [16] mentions the use of various techniques of

XAI in brief. [17] has used two popular techniques namely Sensitivity Analysis (SA) and Layer-wise Relevance Propagation (LRP) for interpreting neural networks. Sensitivity analysis relies on iterative feature occlusion technique, the results of which cannot be computed in a single pass. LRP is based on technique of redistributing the predictions to each layer of neurons until a certain relevance is assigned to each input neuron. This work along with the work in [18] highlights the use of LRP for successfully interpreting deep learning models. [19] introduced a surrogate model called Local Interpretable Model-Agnostic Explanations (LIME) to ensure interpretability and local fidelity of a classifier or a regressor. The published work has successfully demonstrated the flexibility by explaining different models (like random forest) on textual data and non-linear models on image data using this algorithm. [20] introduced game theory based model called SHAP which provides a unified tool for finding feature importance.

This paper presents a novel solution combining techniques to tackle the imbalanced class as well as the black box problem. We do this by applying a resampling technique (SMOTE [10]) on an imbalanced class dataset (Paysim: Financial fraud dataset [1]) to obtain a highly accurate classifier. The classifications are supported by explanations obtained by using XAI techniques (like LIME [19], SHAP [20], LRP [21]) providing interpretability to the classifier.

The remainder of this paper is divided into three sections. In Section 2, we provide a brief review of the performance measures, existing algorithms for classification, resampling and XAI that we have used. The case study with the methodology used and results are presented in Section 3. Finally, in Section 4, we conclude this paper with our observations and some future research directions.

II. PRELIMINARIES

For most datasets Neural Networks (Multilayer Perceptron) and Boosting algorithms (Extreme Gradient Boosting Classifiers) have been known to provide the most accurate models among the current machine learning models. Therefore we use Multilayer Perceptron and XGBoost model to achieve a highly accurate model for our dataset and justify their predictions using various and justify their predictions using various forms of XAI algorithms. These algorithms include: LRP, SHAP and LIME. This section gives a brief working of these algorithms. The performance measures like Confusion Matrix, Precision, Recall, F-Score, False Positive Rate and Area under the curve were used. The brief description of these measures is given in [22].

A. SMOTE

Synthetic Minority Oversampling Technique (SMOTE) is a simple cluster based oversampling technique that has been widely accepted in the community. It chooses 2 samples of the minority class and then assigns it the properties between the two samples. For each of the feature the difference in the value is obtained and then a random

value r is generated such that it lies between 0 and 1, thus the features of the new sample are generated in accordance to the below equation:

$$X_{ni} = \min(X_{ji}, X_{ki}) + |X_{ji} - X_{ki}| \times r \quad (1)$$

where X_{ni} is the i^{th} feature of the n^{th} new sample and X_j, X_k are the random 2 samples of the minority class and $r \in (0,1)$

Thus, a set of oversampled values is obtained. Using this technique the 2 classes are made almost equally proportioned. This method works well when the number of features are less as mentioned in [10]. Thus this simple algorithm ensures that the correlation among the features is conserved even when a large number of synthetically created samples are added to the dataset. Thus this feature allows us to carry out XAI techniques [16] on the dataset obtained from oversampling of the dataset.

B. LRP

Neural network is nothing but a complex function of various input features which return certain predictions. The complex structure with number of hidden layers each with different activation makes it the black box which is difficult to interpret. Layerwise Relevance Propagation is a technique of enabling explainability to neural networks by the principle that inner layer of neurons being connected to outer layers form the basic source of information for the predictions given by the output layer. In other words, LRP works on a principle of information propagation by a conservative property, where whatever is received by a neuron must be redistributed to neurons of outer layers in equal amounts. The output of outer layer in terms of input layer is given by

$$y = f(wx + b) \quad (2)$$

Here w and b are weights and biases of the model respectively and f is the activation function. Explanation provided by LRP is the function of learned representation of the model and the parameters of an instance for which the explanations are to be produced. The relevance score from i_{th} layer to j_{th} layer is given by implementing back propagation in deep neural network. This relevance score from i_{th} layer to j_{th} layer is given by:

$$R_{i \leftarrow j} = \frac{y_i \times w_{ij} + \frac{\epsilon \times \text{sign}(y_j) + \delta \times b_j}{N}}{y_j + \epsilon \times \text{sign}(y_j)} \times R_j \quad (3)$$

This goes on till relevance score of input features (i.e. first layer) is calculated. Here δ and ϵ are constants of the equation. The relevance message transmuted from outermost layer to a layer preceding it is the prediction probabilities of each class. The relevance obtained for the input features is given by:

$$R_i = \sum_j R_{i \leftarrow j} \quad (4)$$

It must be noted that each step of propagation procedure can be modeled as a separate Taylor decomposition over activation outputs of the network [17].

C. SHAP

SHapley Additive exPlanations (SHAP), introduced by [20], is a unified tool for measuring feature importance of various models. It is based on the Shapley Values which are the mean marginal contributions of a given feature value for all the possible combinations of the features. Thus the SHAP values obtained are the Shapley values(S) of the conditional expectation function of the model used and can be calculated as:

$$\varphi_i(p) = \sum_{S \subseteq N/i} \frac{(|S|!(n-|S|-1)!)}{n!} (p(SU_i) - p(S)) \quad (5)$$

The values provide the unique additive feature importance measure that adhere to local accuracy, missingness and consistency. SHAP provides support for both Model-Agnostic explanations as well as Model-Specific explanations. In [20] SHAP has been proven to give explanations consistent with humans and Kernel SHAP also has better computational efficiency than LIME and Shapley values.

D. LIME

Local Interpretable Model-Agnostics Explanations (LIME), introduced by [19], is an implementation of local surrogate models which are used to give explanations for predictions of individual instances of the dataset. LIME does not use the global surrogate model, instead rely on locally trained interpretable models for giving explanations. The underlying methodology of LIME is quite simple. First, a new dataset is chosen out of the desired instances for which explanations of predictions are required. This dataset is then permuted and an interpretable model is trained over it. The weights of this interpretable is the similarity score of these data samples with the original data samples. Both the complex model and the local model are used to make predictions from this local dataset. Thus the local fidelity is calculated which is an approximation of how the model performs locally on these instance. This fidelity is used to give explanation of given instance x which gives how close the explanation is to the predictions of original model.

III. CASE STUDY

A. Overview

We took up the fraud detection dataset [1] which met the characteristics of being highly imbalanced with the proportion of valid and fraud transactions being 99.9% and 0.1% respectively. The dataset comprises of about 6 million records of which only about 8,000 are actually fraud. Thus this imbalanced dataset fed directly to a classifier gave results as characteristic to any imbalanced dataset, a model biased towards the majority class. The dataset had 11

features including one for the target class, 2 being name of sender and recipient. On the given dataset we use the oversampling technique of SMOTE, which has been proven to improve the results, and thus see a significant improvement in reliability of the model. Through Fig.1 we conclude that the oversampling technique does not modify the distribution of the samples. This feature allows us to apply the various explainable AI methods on the given dataset and obtain reliable results with them. Thus in our experiment we have fed the oversampled data to the explainable AI algorithms such as LIME [19], SHAP [20] and LRP [21] to obtain explanations for the way a given sample has been classified.

B. Methodology

The dataset [1] consists of 10 features of which is Flagged (target), is Flaggedfraud, nameOrigin, nameDest were dropped. The dataset did not contain any null values. The TYPE column, which basically represents type of transaction, consisted of categorical values which were one hot encoded into 5 features. Out of these 5 features 3 namely CASH_IN, DEBIT, PAYMENT did not exist in fraud samples. Hence, samples with CASH_OUT and TRANSFER were kept and rest were discarded. Thus the dataset looked as follows: STEP, Amount, OldBalanceOrig (OBO), NewBalanceOrig (NBO), OldBalanceDest (OBD), NewBalanceDest (NBD), type_CASH_OUT, type_TRANSFER. The features Amount, OBO, OBD, NBO, NBD, amount were scaled using a standard scaler. The dataset obtained in step 1 was passed to the SMOTE function to oversample the dataset. An imbalanced sample was passed to smote function which returned a sample with equal number of valid and fraudulent transactions.

C. Result

The results obtained through our experiments are exceedingly intriguing. As far as oversampling by SMOTE is concerned, the correlation representation (i.e. heatmaps) of input features of original and re-sampled datasets as seen in Fig. 1, fairly match with each other. This shows that oversampling technique was successful in eliminating the imbalanced nature of the dataset by producing new fraudulent samples with unaltered feature correlations. Since XAI relies on the features to provide explanations for the predictions, the feature correlation being unaltered was a must. However, the accuracy obtained before and after oversampling varied significantly. The models were tested on the 30% of the original dataset which consist of about 800000 valid samples and about 2500 fraud samples. Two algorithms namely MLP and XGBoost were trained on this dataset. For MLP, we used 3 layers of hidden layers with 8 (equal to number of features) neurons each. For XGBoost and for number of estimators of MLP we used exhaustive parameter search to get the best possible hyperparameters.

From Table 1, we can see that confusion matrix obtained from models of oversampled dataset is far better than that obtained from models of undersampled datasets. Now on a reliable model we can apply the various XAI techniques on these trained models. For the model trained on the oversampled data we have applied the XAI algorithms on a valid and an

invalid sample. Figures 2-4 provide explanation obtained for a valid and fraud samples.

SHAP gives feature importance in descending order to oldbalanceOrg, amount and step for the valid sample while for the invalid sample the feature importance in descending order to oldbalanceOrg, step and amount. Thus SHAP provides almost consistent feature importance for the XGBoost model.

LRP explains the MLP model and for the valid sample the 4 most important features are step, type_CASH_OUT, amount and oldbalanceOrg. For the invalid sample the 4 most important features are oldbalanceOrg, step, amount and newbalanceOrg. Thus we can see that the results seem consistent as 3 of the 4 most important features are common. LIME explains the XGBoost model and for the valid sample the 4 most important features are oldbalanceOrg, newbalanceOrig, step, newbalanceDest. For the invalid sample the 4 most important features are oldbalanceOrg, newbalanceOrg, amount, and type_CASH_OUT. Thus we

can see that the results seem consistent as 3 of the 4 most important features are common.

TABLE I. RESULTS FROM THE CLASSIFIER

	MLP		XGBoost	
	112220 (FN)	716412 (TP)	828618 (TP)	14 (FN)
Undersampling	2262 (TN)	229 (FP)	620 (FP)	1871 (TN)
AUC	0.8633		0.8755	
Oversampling	805738 (TP)	22871 (FN)	793922 (TP)	34687 (FN)
	56 (FP)	2481 (TN)	33 (FP)	2481 (TN)
AUC	0.9723		0.9785	

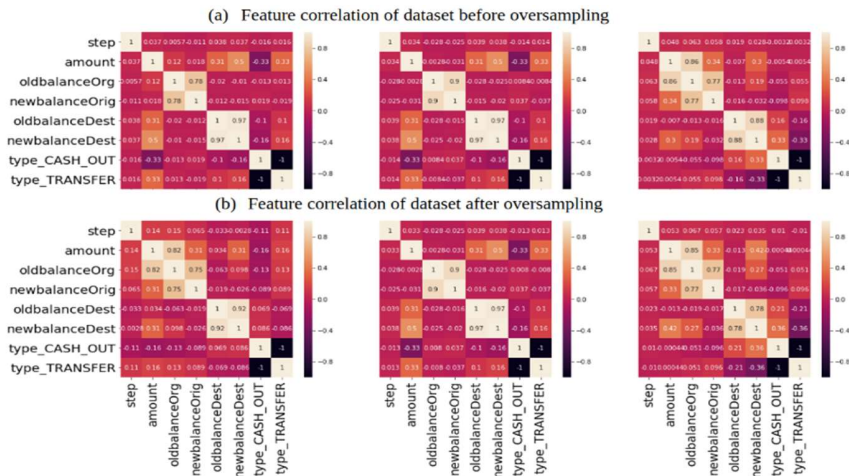


Figure 1. Heatmaps for feature correlation - Whole, valid and fraud samples respectively

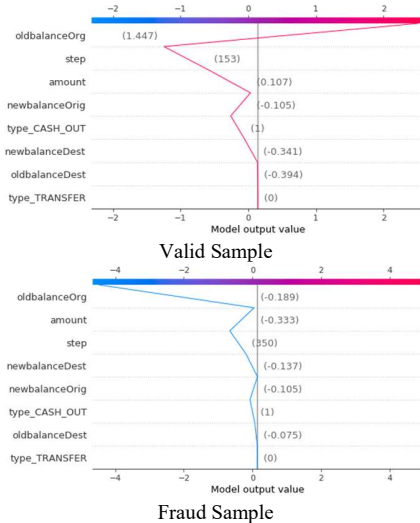


Figure 2. SHAP explanations.

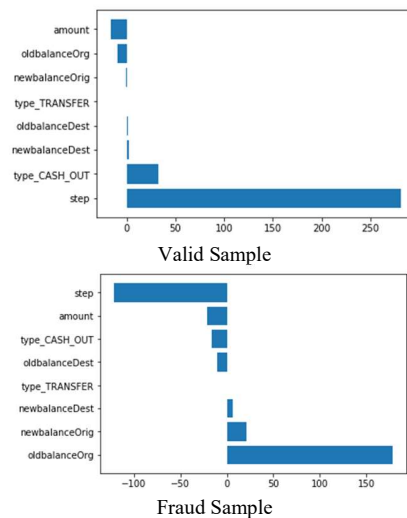


Figure 3. LRP explanations.

Feature	Value	Feature	Value
oldbalanceOrg	-0.19	oldbalanceOrg	1.45
newbalanceOrig	-0.10	newbalanceOrig	-0.10
step	350.00	amount	0.11
newbalanceDest	-0.14	type_CASH_OUT	1.00
amount	-0.33	type_TRANSFER	0.00
oldbalanceDest	-0.08	newbalanceDest	-0.34
type_CASH_OUT	1.00	oldbalanceDest	-0.39
type_TRANSFER	0.00	step	153.00

a. Valid Sample

b. Fraud Sample

Figure 4. Lime explanations.

IV. CONCLUSION

The problem of imbalanced class detection has been an issue faced by the community for quite some time, and many questions have been raised about the reliability of those results. In the present study we approach the imbalanced dataset problem by over sampling the dataset by SMOTE and verifying the reliability of the models and oversampling method with the help of XAI techniques. The results showcase the higher accuracy obtained by the over-sampled dataset while ensuring that the oversampling does not alter the feature correlation of the original dataset. The XAI techniques used comply with each other, by consistently picking out the same features that contributed towards classification of the sample thus justifying the prediction. This approach cultivates trust among service providers and practitioners as the typical black boxes are made more interpretable and explanations for the prediction of each instance helps, provide accountability to such models. This approach however neglects the importance of proper feature engineering of the dataset which is usually carried out on such datasets. The future work of our study is aimed at using this model for high dimensional datasets where exhaustive feature engineering is required to eliminate noise in the dataset.

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